

Abstract: Approximately 7,000 rare diseases exist yet the vast majority lack approved gene therapies, despite the platform potential of AAV and lentiviral vectors. The bottleneck is not biology — it is the lack of a systematic method to identify which existing approved gene-therapy program provides the best biological and regulatory *precedent* for an untreated disease. NanoGT is a Python CLI framework that, given an Orphanet disease identifier, queries public databases, scores 21 approved or clinical-stage gene therapy programs across **14 biologically and clinically relevant dimensions**, and returns a ranked precedent list with confidence classification and plain-language rationale. Validated across a 46-disease proof-of-concept cohort — including positive controls, stress tests, and non-loss-of-function mechanisms — NanoGT correctly recovers known clinical precedents and identifies a systematic gap: many loss-of-function diseases with feasible AAV delivery have *no active therapy programme*, suggesting economics, not biology, is the primary barrier.

BACKGROUND & PROBLEM

The Rare Disease Treatment Gap

~7,000 distinct rare diseases are identified globally. Collectively they affect hundreds of millions of people, yet only a small fraction have disease-modifying treatments. For *nano-rare* diseases (prevalence < 1 in 1,000,000), commercial return is vanishingly small and diseases are systematically deprioritised by industry — leaving patients with no approved therapy, no active trial, and no funded research programme.

Gene Therapy as a Platform Solution

Landmark approvals — **Luxturna** (2017), **Zolgensma** (2019), **Hemgenix** (2022) — established that single-administration gene therapy can produce durable correction across diverse tissue targets. Critically, the foundational regulatory and safety work for one vector-tissue combination becomes a *shared resource*: a subsequent programme using the same vector in the same tissue can leverage this established precedent, dramatically reducing development costs.

The Precedent Problem

For a given untreated disease, which existing approved programme is the best biological and regulatory precedent? This assessment is currently performed manually, is time-consuming, and is inaccessible to small patient groups. No public computational tool integrates the full range of relevant parameters to make this assessment systematically.

METHODS — NANOGT FRAMEWORK

System Architecture

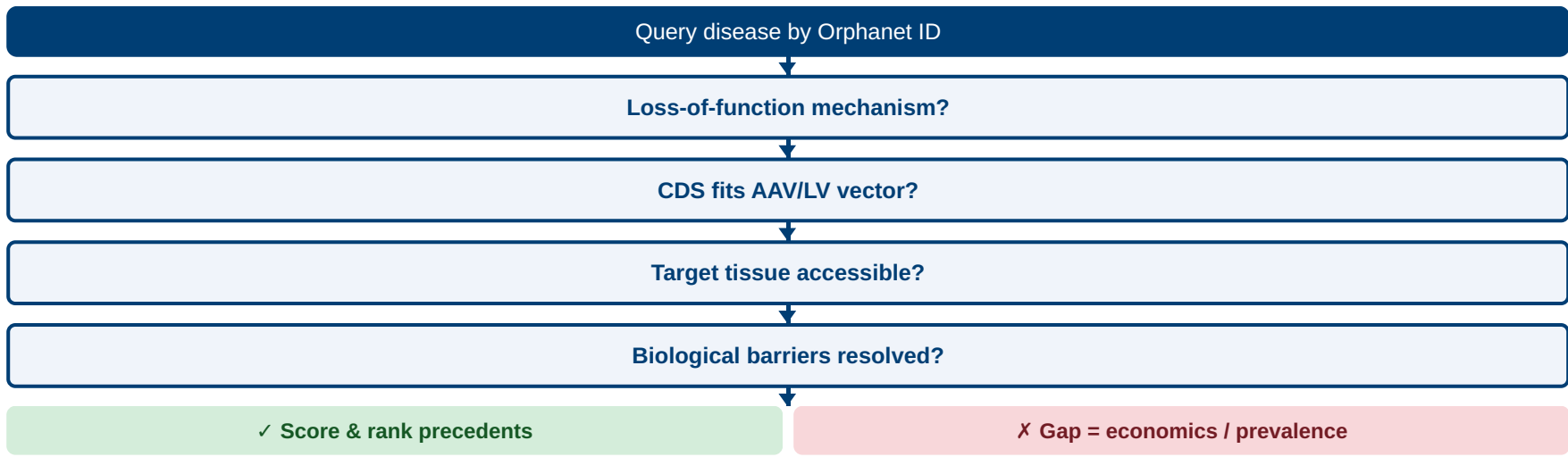
NanoGT is a Python CLI tool (`nanogt match [ORPHA- ID]`) that: (1) queries Orphanet for disease phenotype and causal gene, (2) retrieves gene properties from UniProt, (3) scores 21 programs in a curated surrogate database across 14 dimensions, and (4) outputs a ranked match list with composite scores and a Markdown/ PDF report.

14 Scoring Dimensions

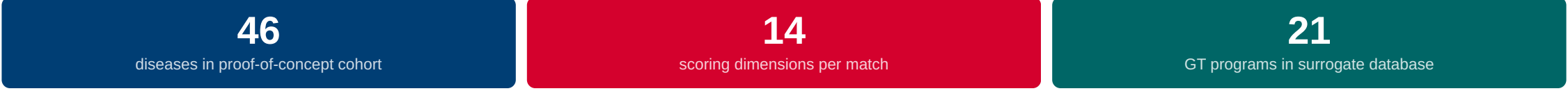
- Gene packaging fit in AAV/LV
- Vector tissue tropism
- Protein class & cross-correction
- Biological pathway similarity
- Mechanism/modality compatibility
- Inheritance pattern
- Regulatory approval stage
- Vector immunogenicity
- Therapeutic window
- Immune privilege of target tissue
- Promoter availability
- Route of administration
- Organelle targeting feasibility
- Cross-correction potential

Raw score normalised to /10. Classified as **high** ≥ 7.5 · **medium** ≥ 5.0 · packaging hard-fail if CDS exceeds vector capacity.

Decision Framework



RESULTS



Positive Controls — Framework Validation

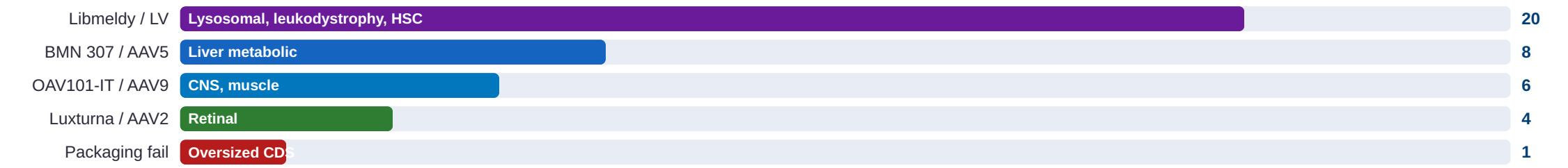
Disease	Gene	Top Precedent	Vector	Score	Confidence
Spinal Muscular Atrophy	SMN1	OAV101-IT (Zolgensma)	AAV9	8.3/10	High ✓
Hemophilia B	F9	Hemgenix	AAV5	9.9/10	High ✓
Hemophilia A	F8	Hemgenix	AAV5	8.5/10	High ✓
ADA-SCID	ADA	Strimvelis	LV	8.1/10	High ✓

✓ **Validation confirmed**
All positive controls recovered their clinically expected precedent as the top-ranked match, validating the 14-dimension scoring logic.

Selected Pilot Cohort Results

Disease	Gene	Top Precedent	Score
MPS type I	IDUA	Libmeldy (LV)	9.5/10
MPS type II	IDS	Libmeldy (LV)	9.3/10
MPS type IIIA (Sanfilippo)	SGSH	Libmeldy (LV)	9.2/10
Metachromatic leukodystrophy	ARSA	Libmeldy (LV)	9.2/10
Gaucher disease	GBA1	Libmeldy (LV)	9.0/10
Krabbe disease	GALC	Libmeldy (LV)	9.0/10
Fabry disease	GLA	Libmeldy (LV)	8.9/10
Duchenne MD (stress test)	DMD	SRP-9001 (AAV9)	7.6/10
Leber hereditary optic neuropathy	MT-ND4	CPCB-RPE1	6.4/10
Neurofibromatosis type 1	NF1	—	Packaging fail
Rett syndrome	MECP2	Skysona (LV)	7.4/10
Crigler-Najjar type I	UGT1A1	Hemgenix (AAV5)	7.2/10

Precedent Clustering by Tissue Target



Stress Tests & Edge Cases

- DMD (oversized cargo):** CDS = 11,055 bp; framework correctly penalises standard AAV and returns micro-dystrophin precedent (SRP-9001) at 7.6/10 — confirming the packaging gate functions.
- LHON (mitochondrial):** MT-ND4 is mitochondrially encoded; allotropic delivery required. Score appropriately reduced to 6.4/10 (medium).
- NF1 (haploinsufficiency, 8.4 kb):** Hard packaging fail — no single-vector precedent survives. Correctly flags dual-vector or non-viral approach needed.

KEY FINDINGS

If a disease has clear loss-of-function biology, an AAV-deliverable gene, and a tractable target tissue — but no gene therapy programme exists — the gap is likely **economics and prevalence**, not biology. NanoGT makes this gap visible and quantifiable.

Biological Findings

- Lysosomal cluster:** Diseases sharing secreted lysosomal enzyme biology (MPS I, II, IIIA, MLD, Krabbe, Gaucher, Fabry) converge on Libmeldy/LV as top precedent — cross-correction biology is a powerful score driver.
- Liver metabolic cluster:** Hepatic enzyme disorders (Crigler-Najjar, OTC deficiency, PKU, methylmalonic acidemia) cluster around BMN 307/AAV5 — established liver tropism and manufacturing precedent transfers broadly.
- Retinal cluster:** Luxturna/AAV2 and CPCB-RPE1/AAV8 capture most retinal LOF diseases — immune-privileged compartment and cone/rod tropism are decisive.
- Membrane proteins score lower:** MCOLN1 (Mucopolidosis IV) and NPC1 cannot exploit cross-correction; per-cell delivery requirement increases complexity and reduces confidence.

Non-LoF Extension

The framework was extended to five non-loss-of-function mechanisms: *haploinsufficiency* (Rett, TSC, NF1, CHARGE), and *repeat-expansion silencing* (Fragile X, Friedrich ataxia). Friedrich ataxia scores 7.9/10 — cDNA addition bypasses the silenced endogenous allele, making gene addition mechanistically appropriate. NF1 fails the packaging gate, correctly flagging a fundamental constraint.

Novel finding — Organelle targeting dimension

Standard frameworks miss a category error: proteins requiring delivery to specific subcellular compartments (mitochondria, peroxisomes) must have intact targeting signals in the therapeutic construct. NanoGT scores this explicitly. Primary hyperoxaluria type 1 (AGXT, a peroxisomal enzyme) and methylmalonic acidemia (MMUT, a mitochondrial matrix enzyme) receive appropriate caveats.

LIMITATIONS

- Surrogate database is curated (21 programs) — not comprehensive. Clinical-stage programmes may be missed.
- Scoring is heuristic, not probabilistic. Weights are biologically motivated but not trained on outcome data.
- Disease biology relies on Orphanet curation and OMIM annotations — errors in source data propagate.
- Framework does not assess manufacturability, immunogenicity at individual patient level, or trial feasibility.
- Not a clinical decision tool — outputs require expert review and disease-specific validation before any therapeutic inference.

CONCLUSIONS

- NanoGT correctly identifies clinically validated precedents for all positive control diseases, confirming the 14-dimension framework is biologically coherent.
- The 46-disease cohort reveals systematic clustering of rare diseases around a small number of approved vector-tissue combinations — the foundational precedent work already done could benefit many untreated diseases.
- Diseases where LoF is clear, AAV delivery is feasible, and no programme exists represent gaps attributable to economics rather than biology — a result meaningful in itself.
- NanoGT provides an open, extensible, public-database-driven tool that makes precedent identification accessible to patient groups and small academic teams.

REFERENCES

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Code: github.com/[repo] · Data: Orphanet · UniProt · OMIM